CPEN 416 / EECE 571O

Lecture 19: Intro to quantum error correction

Tuesday 18 November 2025

Announcements

- Quiz 8 today
- Final week schedule updates
  - Mon. 1 Dec: last tutorial, exam review
  - Tues. 2 Dec: quiz 10, groups 01-02 presentations
  - Thurs. 4 Dec: groups 03-05 presentations

Last time

We defined *quantum channels*,

$$\rho \to \rho' = \Phi(\rho)$$

Quantum channels are CPTP linear maps.

- **T**race-**P**reserving: $\mathrm{Tr}(\rho') = \mathrm{Tr}(\rho)$
- Positive: ρ eigvals $\geq 0 \to$ same ρ'
- **C**ompletely **P**ositive: $I_n \otimes \Phi$ also positive

Last time

Quantum channels are characterized by **Kraus operators** $\{K_i\}$,

$$\Phi(\rho) = \sum_i K_i \rho K_i^\dagger \qquad \sum_i K_i^\dagger K_i = I$$

Example: unitary channel, $\mathcal{U}$, $\{K_i\} = \{U\}$.

$$\mathcal{U}(\rho) = U \rho U^\dagger$$

Example: a projective measurement $\mathcal{M}$, $\{K_i\} = \{\Pi_i\}$

$$\mathcal{M}(\rho) = \sum_i \Pi_i \rho \Pi_i^\dagger$$

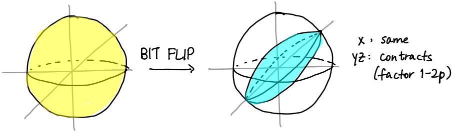
Last time

We expressed a noisy processes as quantum channels.

Bit flip channel

$$\Phi(\rho) = (1-p)\rho + p X \rho X$$

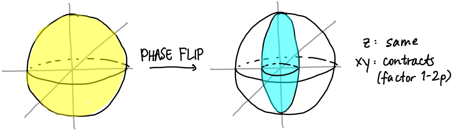
$$K_0 = \sqrt{1-p}\, I$$
$$K_1 = \sqrt{p}\, X$$



Phase flip channel

$$\Phi(\rho) = (1-p)\rho + p \cdot Z \rho Z$$

$$K_0 = \sqrt{1-p}\, I$$
$$K_1 = \sqrt{p}\, Z$$



Last time

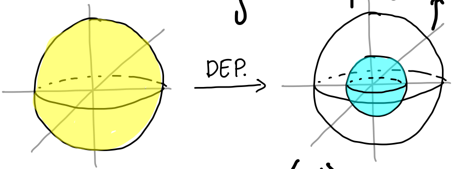
Depolarizing channel

$$\Phi(\rho) = (1-p)\rho + \frac{p}{3}\left(X\rho X + Y\rho Y + Z\rho Z\right)$$

$$\rho = \sum p_i |\psi_i\rangle\langle\psi_i|$$

$H \rho H$

$RX(\theta)\, \rho\, RX^{\dagger}(\theta)$



$$\langle Y \rangle = \mathrm{Tr}(\rho Y)$$

Amplitude damping channel

$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad |+\rangle\langle +| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$1/4$ $\qquad$ $3/4$

Last time

We compared density matrices using two distance measures.

Trace distance

Fidelity

We simulated these processes and computed distance measures using PennyLane's ‘‘`default.mixed`’’ device.

Learning outcomes

Module 5: introduction to quantum error correction

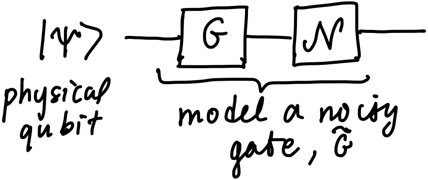
Today: the basics by example

1. Encode quantum information into a logical qubit using a repetition code
2. Design circuitry to correct bit flips and phase flips using repetition codes
3. Correct bit and phase flip errors with the 9-qubit code

Next time: formalizing today's ideas

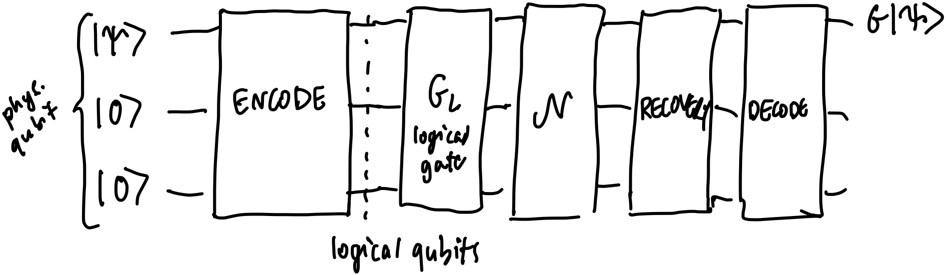
Quantum error correction

Our current picture of noise:



$$\rho = \mathcal{N}\left(G|\psi\rangle\langle\psi|G^\dagger\right)$$

We can protect qubits against noise using error correcting codes.



Repetition codes

Imagine a bit string transmitted through a classical channel that flips each bit (individually) with probability p.

$$\phi\,(0\ 0\ 0\ 1\ 1) \rightarrow 0\ 0\ 1\ 1\ 1$$

Idea: add redundant information to enable *detection* and *correction* of bit flips.

Define *encoding* and *decoding* operations, $\mathcal{E}$ and $\mathcal{D}$,

$\mathcal{E}(0) \rightarrow 000$ "logical 0"

$\mathcal{E}(1) \rightarrow 111$ "logical 1"

$\mathcal{D}(111) \rightarrow 1$

$\mathcal{D}(000) \rightarrow 0$

$0\ 0\ 0\ 1\ 1 \rightarrow 000\ 000\ 000\ 111\ 111$

$000 \downarrow 010$

Repetition codes

Devise a procedure, $\mathcal{R}$, to recover from an error: majority voting.

$$R(b_0, b_1, b_2) = MAJ(b_0, b_1, b_2)$$

Operation	Outcome				
	0	0	1	0	1
$\mathcal{E}$	000	000	111	000	111
ϕ	000	000	101	001	111
$\mathcal{R}$	000	000	111	000	111
$\mathcal{D}$	0	0	1	0	1

Is this better? When will this fail? → more than 1 flip per block

$b_0 b_1 b_2$: $P_e = 3p^2(1-p) + p^3$ if $P_e < P$ ☺

error we can't correct

2 flips ×3

all 3 flipped

Quantum repetition code (bit flip code)

Idea: $\mathcal{E}(|\psi\rangle) \to |\psi\rangle^{\otimes 3}$

Why won't this work?

- no cloning!
- noise is more than bit flips → different for each qubit even!
- measuring to compare them destroys them!

Alternative:

$$\mathcal{E}(|0\rangle) \to |000\rangle = |0\rangle_L$$

$$\mathcal{E}(|1\rangle) \to |111\rangle = |1\rangle_L$$

.

Bit flip code: encoding circuit

Exercise: (a) How does $\mathcal{E}$ affect a general state,

$$\mathcal{E}(\underbrace{\alpha|0\rangle + \beta|1\rangle}_{|\psi\rangle}) \neq |\psi\rangle^{\otimes 3}$$

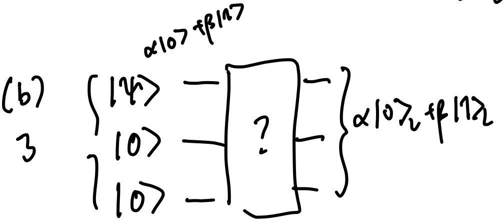
(b) What does the corresponding circuit look like?

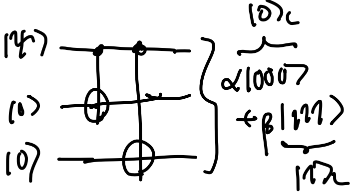
(a) $\mathcal{E}(\alpha|0\rangle + \beta|1\rangle) = \alpha\,\mathcal{E}(|0\rangle) + \beta\,\mathcal{E}(|1\rangle)$

$= \alpha|000\rangle + \beta|111\rangle$

$= \alpha|0\rangle_L + \beta|1\rangle_L$

(b)





Designing logical operations

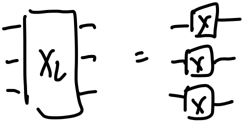
Logical operations are specific to codes.

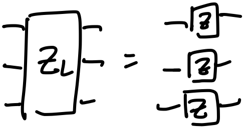
$$\alpha|000\rangle + \beta|111\rangle = \alpha|0\rangle_L + \beta|1\rangle_L$$

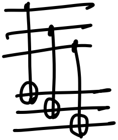
They should act on the logical states $|0\rangle_L$, $|1\rangle_L$ the same way the physical operations act.

$X|0\rangle = |1\rangle \qquad X_L|0\rangle_L = |1\rangle_L$

Exercise: design circuits for logical X, Z, H, S, and $CNOT$.







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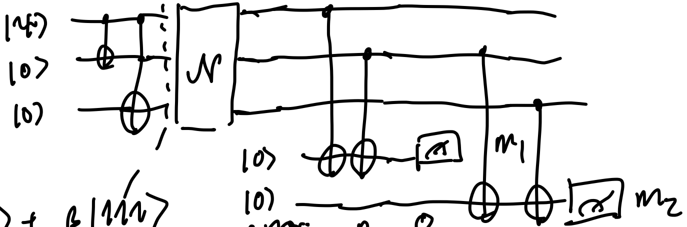
Bit flip code: recovery operation

Let's design a circuit to detect (a) *whether* an error occurs, and (b) *which* error occurs.



$\alpha|000\rangle + \beta|111\rangle$

no error:		0	0
flip qubit 0:	$\alpha\|100\rangle + \beta\|011\rangle$	1	0
flip qubit 1:	$\alpha\|010\rangle + \beta\|101\rangle$	1	1
$\vdots$	$\alpha\|001\rangle + \beta\|110\rangle$	0	1
		m_1	m_2

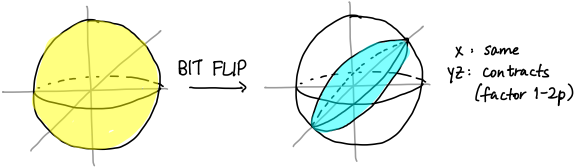
Bit flip code: recovery operation

Design a circuit to *recover* the state of the first qubit

Bit flip code: full circuit for bit flip code

Bit flip code: analysis

Imagine any possible state that can go through the channel:



Exercise: Determine $F(|\psi\rangle, \Phi(|\psi\rangle\langle\psi|))$ where Φ is the *bit flip channel* with parameter p.

Bit flip code: analysis

Compare overall fidelity of the state *after the error*, vs. fidelity *after recovery*.

After the error

After recovery, have the state

Phase flip errors

With our encoding

and appropriate circuitry, we can correct *bit flip errors* but not phase flip errors.

Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

Shor code

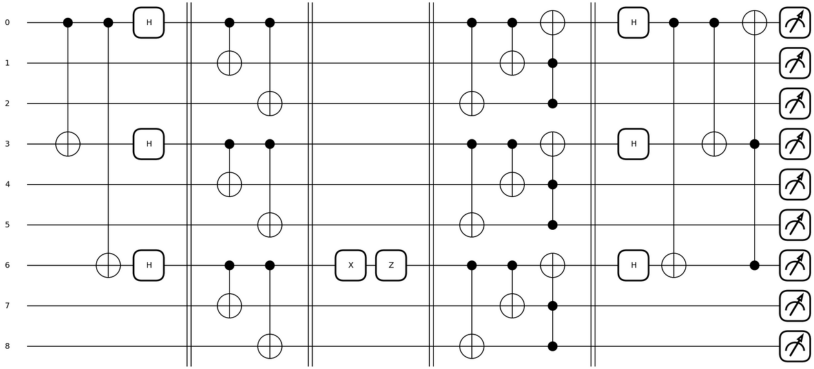
To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

Shor code

The Shor code can correct one *arbitrary* error *on a single qubit*.

Imagine a bit + phase flip on qubit 6:

Shor code



Next time

Next class:

- Properties of errors and error correcting codes
- Stabilizers and stabilizer codes

Action items:

1. Work on project

Recommended reading:

- From this class: Codebook EC; N&C 10.1-10.2
- For next class: Codebook EC; 10.3, 10.5